
Tangent-Admissible Routing for Topology-Constrained Adaptive Variational Quantum Dynamics

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Abstract

Adaptive variational quantum dynamics simulations (AVQDS) grow a pseudo-Trotter circuit according to McLachlan’s variational principle, but the growth rule is blind to the routing cost of the selected two-qubit operators. We introduce tangent-admissible routing (TAR-AVQDS), which couples ansatz growth to a persistent physical layout without changing the McLachlan equations. At every growth event, TAR-AVQDS forms the set of operators whose regularized tangent-space gain is within a fixed fraction of the best available gain, then chooses within that set using the exact immediate shortest-route cost and a Hamiltonian-bond look-ahead potential. A persistent layout, exhaustive eight-qubit initial-layout refinement, and an auditable nearest-neighbor compiler complete the method. We run state-vector simulations for four eight-qubit mixed-field Ising and Lieb–Schultz–Mattis benchmarks on both a 2×4 square lattice and a heavy-hex subgraph. On the square, the topology layer is deliberately inert and incurs $1.00\times$ mean CNOT and $1.00\times$ mean depth. On heavy-hex, it reduces the mean CNOT multiplier from $1.84\times$ for an exhaustive-layout smart naive baseline to $1.57\times$, and depth from $1.80\times$ to $1.46\times$ —mean savings of 13.1% and 16.5%, respectively. All eight required circuits have zero off-topology two-qubit gates and final fidelity at least 0.999143; the smallest margin above the stated accuracy floor is 0.990 percentage points. The method does not win uniformly: on the $h_z = -0.7$ LSM path it uses 2.6% more CNOTs than smart naive routing while reducing depth by 4.4%. This failure, together with route-and-return and layout ablations, identifies when local routing-aware growth is useful and when it is myopic.

1 Introduction

The idea of using a controllable quantum system to simulate another physical system dates to Feynman and underlies universal digital simulation [26, 49, 1, 2, 72]. It now spans quantum chemistry, lattice models, gauge theories, and nonequilibrium many-body dynamics [7, 40, 27, 80, 44, 71, 13, 51, 63]. Fault-tolerant algorithms can organize this work around phase estimation and controlled Hamiltonian simulation; present processors instead reward short, structured circuits [64, 60].

Variational real-time simulation replaces a long product-formula circuit by a parameterized state whose tangent vector approximates Schrödinger evolution [55, 83]. Adaptive variational quantum dynamics simulations (AVQDS) improve this idea by appending Pauli rotations only when the current manifold cannot represent the target tangent to a chosen tolerance [81]. The resulting circuits can be dramatically shorter than first-order Trotter circuits at matched accuracy. The logical circuit, however, assumes arbitrary qubit connectivity. On a sparse device, every nonlocal two-qubit rotation must be exposed by routing, and the added SWAPs may erase much of the logical advantage.

This mismatch is not resolved simply by retaining Hamiltonian-pool operators. The eight-site mixed-field Ising model (MFIM) has a periodic interaction ring, while the Lieb–Schultz–Mattis (LSM)

benchmark has an open interaction chain. The 2×4 square lattice contains a spanning cycle, but the eight-qubit heavy-hex subgraph is a branched tree with neither a Hamiltonian cycle nor a Hamiltonian path. Thus a good static embedding removes all routing from the square benchmark, whereas some routing is unavoidable on heavy-hex. The latter is the scientifically informative regime: the adaptive algorithm chooses among tangent directions, and those directions can have very different physical costs.

We ask whether routing information can be admitted into adaptive growth without replacing the variational principle by a hardware heuristic. Our answer is *tangent-admissible routing* (TAR-AVQDS). The central constraint is that topology may break a near-tie between high-quality tangent directions, but it may not make an arbitrarily poor direction look attractive. The method therefore evaluates the same zero-angle trial append used by AVQDS, computes each candidate’s exact regularized improvement, and admits only candidates within a multiplicative slack η of the best improvement. Routing cost selects inside that set. This gives a crisp separation: McLachlan geometry defines what is admissible; topology decides which admissible direction to take.

The contributions are:

- a fully specified topology-aware AVQDS growth rule based on regularized Schur-complement tangent gain, immediate shortest-route cost, and a persistent-layout interaction potential;
- a modular compiler and evidence trail that account for every logical CNOT and every three-CNOT SWAP, schedule primitive depth, and verify every two-qubit pair against the declared edge set at every growth prefix;
- complete deterministic simulations of four benchmark dynamics on two contrasting sparse topologies, with a stronger-than-static smart naive comparison obtained by exhaustive search over all $8!$ initial layouts; and
- mechanism, sensitivity, compute-cost, and failure analyses, plus a third-topology evaluation on a linear chain.

The contribution is not a new generic router. Rather, it is the use of routing cost *inside* the variational tangent-selection step, while retaining the original equations of motion. This distinction lets us attribute changes in physical cost and fidelity directly to topology-aware ansatz growth.

2 Related work

Adaptive variational dynamics. Yao et al. introduced the append-only pseudo-Trotter AVQDS construction used here [81]. Its later compressed-circuit variant, sometimes called AVQDS(T), appends disjoint operators in TETRIS-like layers and reduces measurement and depth costs [84]. That work exploits parallel support but does not make physical layout a term in the McLachlan candidate decision. Hardware-efficient time-evolution ansätze instead fix connectivity-compatible circuit families in advance [10]; this avoids routing by construction but gives up the Hamiltonian-pool adaptive manifold studied here.

Routing and layout. Generic compilers such as SABRE use heuristic forward/backward search to insert SWAPs and evolve a logical-to-physical mapping [45]. Exact and heuristic formulations of allocation and routing have also characterized the combinatorial problem [69, 19, 85, 59, 16, 70, 36]. Heavy-hex itself is motivated by low-degree fault-tolerant layouts [14]. We borrow the standard notions of a persistent mapping and shortest-path moves, not a published router wholesale: at eight qubits we enumerate all initial layouts and all shortest paths, making the stated routing policy deterministic and exact within its move class. More importantly, a post-hoc router cannot revisit which variational tangent generated a logical gate. TAR-AVQDS exposes that choice before the ansatz is committed.

Connectivity-aware adaptive ansätze. Qubit-ADAPT-VQE constructs shallower adaptive ground-state ansätze from a qubit operator pool [74]; TETRIS-ADAPT-VQE selects disjoint high-gradient operators to increase circuit density [4]. PermVQE uses state correlations to select a qubit permutation for a fixed hardware-efficient VQE ansatz [76]. These methods motivate hardware-aware operator selection or layout, but address energy minimization rather than McLachlan-controlled real-time

evolution. TAR-AVQDS's novel unit is the *near-optimal regularized tangent-gain set*: it limits hardware bias using the same local dynamical objective that triggers AVQDS growth.

Broader variational context. VQE established the hybrid variational pattern [62, 54, 61]; subsequent work developed hardware-efficient, coupled-cluster, and adaptive constructions [39, 66, 9, 65, 31]. The same landscape includes QAOA [25], derivative and metric-estimation methods [57, 50, 56], and analyses of trainability [53, 30, 6]. Real-time alternatives include general variational simulation, variational fast forwarding, variational Hamiltonian diagonalization, and projected variational dynamics [24, 17, 18, 15, 8, 48]. These methods clarify the design space but do not combine a McLachlan candidate certificate with persistent topology cost.

3 Methodology

3.1 AVQDS geometry retained unchanged

Let the ordered pseudo-Trotter ansatz be

$$|\psi(\theta)\rangle = U_M(\theta_M) \cdots U_1(\theta_1)|\psi_0\rangle, \quad U_\mu(\theta_\mu) = \exp(-i\theta_\mu P_\mu/2), \quad (1)$$

where P_μ is a one- or two-qubit Pauli string in the Hamiltonian pool. The factor 1/2 follows the released implementation. Appending a rotation at $\theta = 0$ leaves the state invariant and enlarges its tangent space. The underlying geometry is the projective-state metric [3, 42, 56], rather than an arbitrary circuit score.

Writing $|\partial_\mu\psi\rangle = \partial|\psi\rangle/\partial\theta_\mu$, the density-matrix form of McLachlan's principle used by the released code is

$$M_{\mu\nu} = \text{Re}[\langle\partial_\mu\psi|\partial_\nu\psi\rangle + \langle\partial_\mu\psi|\psi\rangle\langle\partial_\nu\psi|\psi\rangle], \quad (2)$$

$$V_\mu = \text{Im}[\langle\partial_\mu\psi|H|\psi\rangle - \langle\partial_\mu\psi|\psi\rangle\langle H\rangle], \quad (M + \xi I)\dot{\theta} = V, \quad (3)$$

with $\xi = 10^{-6}$. In the code-normalized convention, the residual squared distance is

$$\mathcal{D}^2 = \text{Var}_\psi(H) - V^\top(M + \xi I)^{-1}V. \quad (4)$$

This is one half of a convention sometimes used in prose descriptions of AVQDS; every threshold and logged value here uses Eq. (4), lower-clipped at zero for roundoff. When $\mathcal{D}^2 \geq 10^{-3}$, operators are trial-appended at zero angle until the threshold is restored. Parameters then take a forward-Euler step with $\max_\mu |\delta\theta_\mu| \leq 0.005$ and $\delta t \leq 0.05$. The conditioning cutoff is $\max_\mu |\dot{\theta}_\mu| < 7000$. These equations, regularization, pool ordering, append-only policy, and integrator are identical for unconstrained AVQDS and TAR-AVQDS.

3.2 Regularized tangent gain

The hardware decision uses the exact local improvement from a candidate, not a proxy such as support distance. Let $R = M + \xi I$, $q = V^\top R^{-1}V$, and consider zero-angle candidate a . Its enlarged regularized system is

$$\tilde{R}_a = \begin{bmatrix} R & b_a \\ b_a^\top & \alpha_a \end{bmatrix}, \quad \tilde{V}_a = \begin{bmatrix} V \\ v_a \end{bmatrix}. \quad (5)$$

Define the Schur denominator and residual

$$s_a = \alpha_a - b_a^\top R^{-1}b_a, \quad r_a = v_a - b_a^\top R^{-1}V. \quad (6)$$

Then the additional captured tangent norm is exactly

$$g_a = \tilde{V}_a^\top \tilde{R}_a^{-1} \tilde{V}_a - q = \frac{r_a^2}{s_a}, \quad \mathcal{D}_a^2 = \text{Var}(H) - q - g_a. \quad (7)$$

Numerically, s_a is lower-clipped at 10^{-14} before division. All candidates share the same factorization of R . Thus the Schur form algebraically reproduces an explicit enlarged solve while substantially reducing classical work. An exhaustive candidate test found a maximum captured-norm discrepancy of 1.78×10^{-15} relative to explicit solves; finite-difference derivatives agree to 1.61×10^{-10} .

For reference, the released choice scans the fixed pool order with running values initialized to q and $\max(\max_{\mu} |\dot{\theta}_{\mu}|, 7000)$. A trial replaces the running choice if its captured norm exceeds the running norm by 10^{-8} , or if its distance is below 10^{-3} and its largest velocity is smaller than the running velocity. The last qualifying candidate is retained; an immediate repeat of the current final operator is skipped. This rule is used unchanged for the unconstrained baseline and for TAR-AVQDS's equal-route-cost guard.

Let $\mathcal{C}$ exclude only an immediately repeated copy of the last operator, as in the released selection rule, and let $g_{\star} = \max_{a \in \mathcal{C}} g_a$. TAR-AVQDS admits

$$\mathcal{S}_{\eta} = \{a \in \mathcal{C} : g_a \geq (1 - \eta)g_{\star}\}. \quad (8)$$

The implementation allows 10^{-14} absolute tolerance in this comparison. We use $\eta = 0.1$, selected once on the MFIM $h_z = 0$ heavy-hex calibration and then frozen. Equation (8) is the safeguard against a cheap but dynamically irrelevant operator: topology can trade at most the declared fraction of the best instantaneous gain.

3.3 Persistent routing state and candidate choice

A mapping $m : \{0, \dots, 7\}_{\text{logical}} \rightarrow \{0, \dots, 7\}_{\text{physical}}$ is a permutation. For topology graph G , let d_G denote shortest-path distance and let $\mathcal{E}_H$ be the distinct logical interaction bonds of the simulated Hamiltonian. We use the look-ahead potential

$$\Phi(m) = \sum_{(i,j) \in \mathcal{E}_H} [d_G(m(i), m(j)) - 1]. \quad (9)$$

It is zero exactly when every Hamiltonian bond is local under m . Before growth, all $8!$ mappings are enumerated and the lexicographically first minimizer of Φ initializes a *shadow* persistent router.

For candidate pair (i, j) at physical distance d , we enumerate every shortest path $(p_0, \dots, p_d)$ and every split $k \in \{0, \dots, d-1\}$. The logical operand at p_0 is moved along the path to p_k , and the operand at p_d is moved backward to p_{k+1} . Every plan uses exactly $d-1$ adjacent SWAPs. Among these shortest plans we select the one with minimum post-move Φ , breaking ties by the SWAP sequence and mapping. The mapping is retained; it is never restored after a rotation. Single-qubit candidates require zero SWAPs and do not change the mapping.

For each candidate a , let s_a^{route} be this immediate SWAP count and m'_a its post-route mapping. If candidates have unequal immediate costs, TAR-AVQDS chooses lexicographically

$$a_{\text{TAR}} = \arg \min_{a \in \mathcal{S}_{\eta}} \left(s_a^{\text{route}}, \Phi(m'_a), -g_a, \max_{\mu} |\dot{\theta}_{\mu}|, \text{index}(a) \right). \quad (10)$$

The chosen route updates the shadow mapping before the next growth decision. If every valid candidate has the same s_a^{route} , the original AVQDS conditioning-aware selection is used exactly. This guard makes topology awareness inert when routing gives no preference, notably for the Hamiltonian-cycle/path embeddings on topology A. It also provides a single switch: disabling Eq. (10) recovers the unconstrained baseline.

3.4 Final compilation, depth, and compliance

The shadow mapping influences growth but is not privileged in final accounting. Once a logical ansatz is fixed, we compile it from every one of the $8! = 40320$ initial layouts using the same persistent shortest-route policy. We choose the layout lexicographically by (total SWAPs, primitive depth, layout). The identical exhaustive refinement is applied to smart naive routing of the unconstrained ansatz. This is stronger than choosing a mapping from Hamiltonian-bond coverage alone and prevents layout quality from artificially favoring TAR-AVQDS.

A two-qubit Pauli rotation is basis changes followed by CNOT- R_z -CNOT and inverse basis changes, hence two CNOTs. A SWAP is three adjacent CNOTs. Consequently,

$$N_{\text{cx}} = 2N_{\theta, 2q} + 3N_{\text{swap}}. \quad (11)$$

Primitive depth is an ASAP schedule of all one-qubit basis/rotation gates and CNOTs in the emitted order: a gate occupies one layer after the latest previous gate on any of its operands. We exclude

Algorithm 1 TAR-AVQDS growth at one integration time

Require: current ansatz, parameters, persistent mapping m , topology G

```
1: while  $D^2 \geq 10^{-3}$  or conditioning cutoff is violated do
2:   factor  $R = M + 10^{-6}I$  and evaluate  $(g_a, \dot{\theta}_a)$  for every zero-angle trial using Eq. (7)
3:   compute the original AVQDS choice  $a_{AV}$ 
4:   for every valid two-qubit candidate  $a$  do
5:     enumerate shortest routes and retain  $\arg \min(\Phi(m'_a), \text{route}, m'_a)$ 
6:   end for
7:   if all immediate route costs are equal then
8:      $a \leftarrow a_{AV}$ 
9:   else
10:    form  $\mathcal{S}_{0,1}$  and choose  $a$  by Eq. (10)
11:   end if
12:   append  $e^{-i\theta_a P_a/2}$  at  $\theta_a = 0$  and update  $m$  with its selected route
13: end while
14: take the unchanged adaptive forward-Euler step
```

initial-state preparation, matching the original benchmark convention. After compilation, every two-qubit primitive is recorded with its physical pair, source, ansatz index, and growth step. A separate audit compares each pair with $E(G)$ and stores cumulative per-step offsets; no logical nonlocal gate is hidden in the count. This gate convention is the standard Pauli-rotation decomposition associated with product-formula simulation [77, 60].

4 Experimental setup

Models. The eight-qubit MFIM is

$$H_{\text{MFIM}} = - \sum_{i=0}^7 Z_i Z_{i+1} - 2 \sum_{i=0}^7 X_i + h_z \sum_{i=0}^7 Z_i, \quad (12)$$

with periodic $i+1 \bmod 8$, $h_z \in \{0, 0.5\}$, initial state $|0\rangle^{\otimes 8}$, a sudden quench at $t = 0$, and final time 3. Its Hamiltonian pools contain respectively 16 and 24 strings, ordered as $(X_0, \dots, X_7)$, then $(Z_0 Z_1, \dots, Z_6 Z_7, Z_7 Z_0)$, and, when $h_z \neq 0$, $(Z_0, \dots, Z_7)$. The open-boundary LSM chain is

$$H_{\text{LSM}}(t) = - \sum_{i=0}^6 [(1 + \gamma(t)) X_i X_{i+1} + (1 - \gamma(t)) Y_i Y_{i+1}] + h_z \sum_{i=0}^7 Z_i, \quad (13)$$

where $h_z \in \{-0.7, 1.6\}$, $\gamma(t) = 1 - 2t/3$ for $0 \leq t \leq 3$, and $\gamma = -1$ for $3 < t \leq 6$. The initial state is the exact ground state at $\gamma = 1$. The pool is ordered $Z_0, \dots, Z_7$, followed by $(X_i X_{i+1}, Y_i Y_{i+1})$ interleaved for $i = 0, \dots, 6$. The model traces to the exactly soluble spin-chain construction of Lieb, Schultz, and Mattis [47]; the MFIM trajectories also probe the dynamical-transition setting characterized through the Loschmidt echo [34, 33].

Topologies and comparisons. Figure 1 gives the exact labeled graphs. Topology A is the ten-edge square; topology B is the seven-edge heavy-hex subgraph. Topology C, an eight-qubit line, is an optional generalization test. In physical-qubit labels,

$$E_A = \{01, 12, 23, 45, 56, 67, 04, 15, 26, 37\}, \\ E_B = \{01, 12, 13, 35, 45, 56, 67\}, \quad E_C = \{01, 12, 23, 34, 45, 56, 67\},$$

where ij denotes the undirected pair (i, j) . We compare (i) logical unconstrained AVQDS, (ii) the same unconstrained ansatz with exhaustive-layout persistent shortest-path routing (“smart naive”), and (iii) TAR-AVQDS with the same final compiler. The primary accuracy criterion is $f_{\text{TAR}}(T) \geq f_{\text{AV}}(T) - 0.01$. We report final fidelity, physical CNOT count, primitive depth, and N_θ ; CNOT and depth multipliers use the paired unconstrained logical circuit as denominator. Exact state evolution uses eigendecomposition for the time-independent MFIM and DOP853 integration for LSM with relative/absolute tolerances $10^{-11}/10^{-12}$ and maximum step 0.01. Repeating the LSM final reference with tolerances $2 \times 10^{-13}/2 \times 10^{-14}$ and maximum step 0.0025 changed fidelity by at most 2.22×10^{-16} . The convergence check follows standard adaptive-integration practice [35, 73].

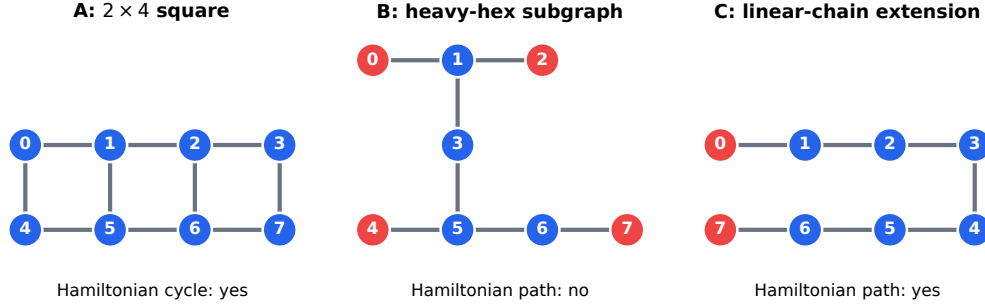


Figure 1: **Evaluated physical graphs.** Node labels are physical qubits and drawn links are the complete allowed edge sets. The square admits a spanning cycle; the branched heavy-hex subgraph admits no spanning path; the line is the optional third topology. Red nodes have degree one.

Calibration and determinism. Only η is new. We evaluated $\eta \in \{0.1, 0.2, 0.4, 0.6\}$ on MFIM $h_z = 0$, topology B, and minimized the geometric mean of CNOT and depth ratios to that configuration’s smart naive circuit, conditional on satisfying the accuracy floor. We then froze $\eta = 0.1$ for every other run. Pool order, layout tie breaks, shortest paths, eigensolver phase, and floating-point execution are deterministic; there are no random seeds or stochastic initializations to average. Every configured run is reported. Simulations used Python 3.14.4, NumPy 2.5.1, SciPy 1.18.0, and one OpenBLAS/OMP thread. Wall time measures the adaptive state-vector core; routing time is reported separately.

Released-code validation. We inspected the complete supplied source archive and vendored its core files unchanged. The optimized Pauli backend retains its density-matrix geometry, global-phase terms, trial-append rule, regularization, Euler stepping, and canonical pool membership. It uses the explicit logical-site order stated above; the release scans the periodic 70 MFIM bond first, so sequential tie-breaking need not be bit-identical. Besides the algebra tests in Section 3.2, we directly ran the released `ansatz.one_step` implementation on all four full benchmarks through the stated final times. The wrapper restores the removed NumPy aliases `complex` and `int`. To avoid repeated dense matrix reconstruction, it applies pool rotations with the exact Pauli identity $e^{\alpha P} = \cosh(\alpha)I + \sinh(\alpha)P$; every pool action is checked against the released spectral formula before each run. The maximum discrepancy over the four full-run checks is 4.29×10^{-17} . The authors’ files and the adaptive class logic remain unchanged. Section 5.1 reports both validation layers.

5 Experimental results

5.1 Unconstrained baseline and source validation

Table 1 shows the unconstrained runs used as the comparison states. Both MFIM fidelities exceed the 99.5% baseline criterion. Logical counts are close to the published reference despite the stricter 0.005 parameter-step cap and deterministic pool tie ordering. The LSM evolutions continue through the full post-ramp interval to $t = 6$, rather than stopping at the end of the ramp.

Direct execution of the released adaptive class provides an independent implementation check (Table 2). Its minimum MFIM final fidelity is 0.999081. Differences in `ansatz` length are expected: the release uses dense QuTiP objects and its conditioning-aware sequential trial rule with the periodic bond first, while the port shares a factorization and evaluates algebraically equivalent trial systems in the logical order stated above. The topology comparisons use the port for both methods, so candidate arithmetic, integration, and exact references are paired.

Table 1: **Unconstrained AVQDS baselines generated by the optimized, algebra-checked port.** $f(T)$ and $\min_t f$ are state fidelities against exact evolution. “paper” is the logical CNOT reference quoted for the original implementation. Wall time is the adaptive core, excluding exact reference generation and routing.

Benchmark	$f(T)$	$\min_t f$	N_{cx}	paper	depth	N_θ	wall (s)
MFIM $h_z = 0$	0.999494	0.999494	132	134	82	126	47.4
MFIM $h_z = 0.5$	0.999244	0.999244	232	210	180	284	138.1
LSM $h_z = -0.7$	0.999572	0.999571	120	106	191	72	48.4
LSM $h_z = 1.6$	0.999199	0.999199	118	106	190	74	55.1

Table 2: **Full direct runs through the supplied authors’ classes.** The 10^{-6} regularizer, 0.005 parameter-step cap, 10^{-3} code-normalized distance cutoff, and exact final times are used. Counts use two CNOTs per two-qubit ansatz parameter.

Benchmark	$f(T)$	N_{cx}	N_θ	wall (s)
MFIM $h_z = 0$	0.999611	136	129	791.4
MFIM $h_z = 0.5$	0.999081	212	280	1644.4
LSM $h_z = -0.7$	0.999648	116	75	836.3
LSM $h_z = 1.6$	0.999619	124	75	1160.8

5.2 Accuracy, topology compliance, and physical cost

All eight TAR-AVQDS circuits meet the accuracy floor and contain no off-topology two-qubit primitive. The least final fidelity is 0.999143, the largest absolute drop from its paired unconstrained state is only 0.010 percentage points, and the least margin above the allowed floor is 0.990 percentage points. The independent gate-stream audit found 0 violations. Tables 3 and 4 report the two model families separately; Table 5 gives all per-instance multipliers and margins over smart naive routing.

On topology A, both interaction graphs embed locally and all candidates have equal immediate routing cost. The guard in Eq. (10) therefore recovers the logical baseline exactly: zero SWAPs, identical ansatz, fidelity, CNOT count, and depth. This is an important negative control. A topology-aware rule should not perturb dynamics when the topology offers no meaningful cost distinction.

On topology B, smart naive routing averages $1.84\times$ the logical CNOT count and $1.80\times$ the logical primitive depth. TAR-AVQDS lowers those averages to $1.57\times$ and $1.46\times$, saving 13.1% CNOTs and 16.5% depth relative to smart naive routing. The improvement is not the artifact of a weak initial mapping: both methods search all 40320 layouts for the complete realized ansatz.

Figure 3 shows that agreement is not confined to the final time. Both the baseline and topology-aware trajectories remain close to exact evolution throughout the quenches. The horizontal line is an infidelity reference; the acceptance criterion compares final fidelity with the paired baseline.

6 Analysis

6.1 Mechanism analysis: fewer costly tangent directions

Equation (11) separates two mechanisms: the number of selected two-qubit rotations and the SWAPs needed to expose them. Figure 4 gives both components. On heavy-hex, TAR-AVQDS changes 60, 164, 29, and 25 growth decisions for MFIM $h_z = 0$, MFIM $h_z = 0.5$, LSM $h_z = -0.7$, and LSM $h_z = 1.6$, respectively. The resulting two-qubit parameter counts fall from 66 to 53, 116 to 90, 60 to 53, and 59 to 50. This is not operator deletion—the ansatz remains append-only. It is a different path through a sequence of locally near-equivalent tangent directions. Such time-dependent circuit growth is qualitatively consistent with the ballistic production of entanglement after quenches and with circuit-complexity growth, although the present data do not infer a scaling law [21, 41, 11].

Table 3: **MFIM final-state and circuit results.** Smart naive routing preserves the unconstrained state exactly and changes only physical circuit cost. Topology A is the square and B is heavy-hex.

Method	topology	MFIM $h_z = 0$			MFIM $h_z = 0.5$		
		fidelity	CNOT	depth	fidelity	CNOT	depth
Unconstrained AVQDS	all-to-all	0.999494	132	82	0.999244	232	180
Smart-embedding naive routing	A	0.999494	132	82	0.999244	232	180
Smart-embedding naive routing	B	0.999494	300	197	0.999244	544	428
TAR-AVQDS	A	0.999494	132	82	0.999244	232	180
TAR-AVQDS	B	0.999396	244	148	0.999143	441	331

Table 4: **LSM final-state and circuit results.** Simulation includes the ramp $0 \leq t \leq 3$ and post-ramp evolution through $t = 6$. State preparation is excluded consistently for all methods.

Method	topology	LSM $h_z = -0.7$			LSM $h_z = 1.6$		
		fidelity	CNOT	depth	fidelity	CNOT	depth
Unconstrained AVQDS	all-to-all	0.999572	120	191	0.999199	118	190
Smart-embedding naive routing	A	0.999572	120	191	0.999199	118	190
Smart-embedding naive routing	B	0.999572	156	228	0.999199	172	230
TAR-AVQDS	A	0.999572	120	191	0.999199	118	190
TAR-AVQDS	B	0.999704	160	218	0.999601	142	198

Routing then changes SWAP counts from smart naive values 56, 104, 12, 18 to 46, 87, 18, 14. Thus the two MFIM wins combine fewer logical entanglers with fewer moves. The $h_z = 1.6$ LSM win has the same structure. The $h_z = -0.7$ case reduces logical entanglers but increases SWAPs enough to reverse CNOT savings; Section 6.2 examines it.

The persistent-layout and final-layout stages are independently useful. Restoring the mapping after every rotation (route-and-return) would require 592, 582, 274, 190 CNOTs across the four heavy-hex cases. Keeping the growth-time initial layout but omitting final exhaustive refinement gives 301, 459, 172, 142; final refinement produces 244, 441, 160, 142. The first ablation isolates the benefit of retaining useful permutations, and the second shows that an online Hamiltonian-bond potential is not a replacement for whole-sequence layout optimization.

The structural potential in Eq. (9) predicts the negative control. Its minimum is zero for both ring and chain on the square. On heavy-hex it is positive because the graph has four degree-one constraints and cannot host either interaction graph as a spanning path/cycle. The line has zero potential for LSM but not MFIM. This graph-level fact correctly predicts where the selector will be active before any dynamics is run.

6.2 Failure analysis: the ferromagnetic LSM path

TAR-AVQDS is not Pareto-improving on LSM $h_z = -0.7$. It produces fidelity 0.999704, higher than the baseline’s 0.999572, and lowers primitive depth from the smart naive value 228 to 218. Yet its CNOT count rises from 156 to 160 (2.6%). The logical part falls by 14 CNOTs, but the persistent trajectory uses six more SWAPs, adding 18 CNOTs. The immediate-cost first term in Eq. (10) cannot see that a sequence of individually cheap moves may strand future LSM bonds across the two sides of heavy-hex’s degree-three junction.

This result delineates the method’s scope. The Hamiltonian-bond potential is a one-step surrogate, not a whole-future ansatz objective. It works especially well when repeated growth choices create a stable bias, as in the longer MFIM ansätze. In the $h_z = -0.7$ LSM trajectory, only 29 choices differ and later operator order matters enough for the local surrogate to lose in SWAP count. The fact that depth still improves is consistent with the scheduler: the changed gate order exposes more parallel single-qubit and disjoint-edge work, whereas the scalar CNOT metric prices every SWAP equally. A beam search over several future admissible growth events could address this failure, but would couple branches to distinct variational states and greatly increase classical cost; it is outside the present method.

Table 5: **Topology overhead for every model and graph.** Multipliers use the paired unconstrained logical CNOT count or depth as denominator. Savings are TAR-AVQDS relative to smart naive routing; a negative CNOT saving records an honest underperformance.

Benchmark	topology	CNOT multiplier		depth multiplier		TAR saving	
		naive	TAR	naive	TAR	CNOT	depth
MFIM $h_z = 0$	A	1.00	1.00	1.00	1.00	+0.0%	+0.0%
MFIM $h_z = 0$	B	2.27	1.85	2.40	1.80	+18.7%	+24.9%
MFIM $h_z = 0.5$	A	1.00	1.00	1.00	1.00	+0.0%	+0.0%
MFIM $h_z = 0.5$	B	2.34	1.90	2.38	1.84	+18.9%	+22.7%
LSM $h_z = -0.7$	A	1.00	1.00	1.00	1.00	+0.0%	+0.0%
LSM $h_z = -0.7$	B	1.30	1.33	1.19	1.14	-2.6%	+4.4%
LSM $h_z = 1.6$	A	1.00	1.00	1.00	1.00	+0.0%	+0.0%
LSM $h_z = 1.6$	B	1.46	1.20	1.21	1.04	+17.4%	+13.9%

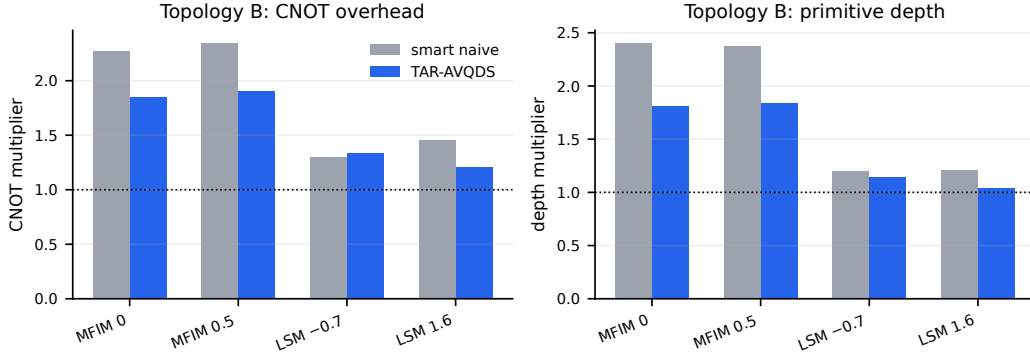


Figure 2: **Heavy-hex overhead by benchmark.** Multipliers are relative to the paired unconstrained logical circuit. TAR-AVQDS reduces both metrics in all four cases, except that its CNOT count is slightly higher on LSM $h_z = -0.7$. The dotted line denotes no topology overhead.

Topology A is a different “failure to improve”: TAR-AVQDS equals rather than beats smart naive routing. This is intentional. There is no routing overhead to remove, and using hardware information to perturb the selected tangent would only introduce avoidable fidelity risk.

6.3 Computational cost trade-off

Table 8 reports paired core and final-routing times. TAR-AVQDS reuses the factorization already needed for trial appends; the extra online work is enumeration of shortest paths on an eight-node graph and tuple comparisons. Its core-time ratios are therefore governed mainly by the number of Euler steps and the realized ansatz size, not by a second dense McLachlan solve per candidate. The routing time ranges from about 10 to 30 seconds on topology B and is dominated by the deliberately exact 8! final-layout enumeration. At larger sizes, that enumeration would be replaced by a common heuristic for both methods; the online selector itself does not require exhaustive layouts.

The comparison is intentionally a scientific cost analysis rather than a hardware runtime prediction. State-vector construction of the full Jacobian scales exponentially in qubit number, and the direct released QuTiP implementation is much slower than the vectorized Pauli backend. Neither classical time should be conflated with quantum shot cost: this study uses exact states and no sampling.

Table 6: **Mean overhead and margin over smart naive routing.** Averages give equal weight to the four benchmark configurations. Topology C is the optional linear-chain extension.

Topology	CNOT multiplier		depth multiplier		TAR saving vs. naive	
	naive	TAR	naive	TAR	CNOT	depth
A	1.00	1.00	1.00	1.00	0.0%	0.0%
B	1.84	1.57	1.80	1.46	13.1%	16.5%
C	1.69	1.50	1.57	1.39	8.0%	8.1%

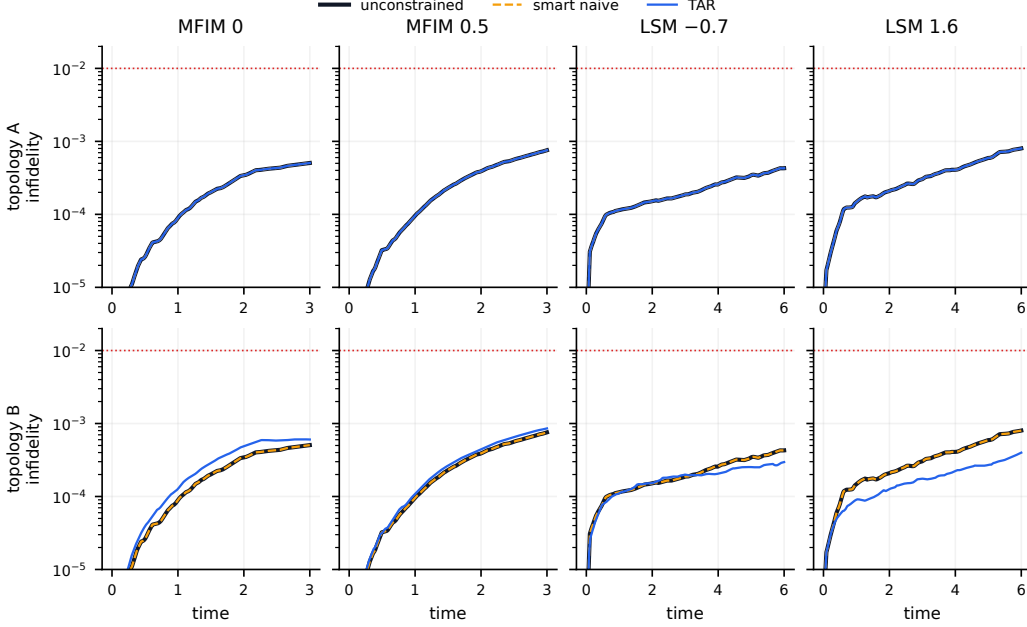


Figure 3: **Time-resolved state infidelity for all eight benchmark-topology pairs.** Smart naive and unconstrained curves coincide because routing is an exact logical relabeling; both are drawn. On topology A, TAR-AVQDS also coincides with them. Differences on topology B arise from changed growth decisions, not SWAP error.

6.4 Sensitivity of the tangent slack

Table 9 reports every calibration value. All four satisfy the accuracy floor. Although $\eta = 0.6$ obtains five fewer CNOTs than $\eta = 0.1$, its depth is 17 layers larger; $\eta = 0.1$ has the lowest declared joint score and was fixed before the remaining runs. The limited spread in fidelity supports the intended interpretation of $\mathcal{S}_\eta$ as a near-tangent-equivalence set, while the nonmonotonic costs warn against interpreting a larger slack as uniformly better.

6.5 Third-topology generalization

Without retuning, we repeat all four benchmarks on the linear chain topology C. The two LSM cases recover the unconstrained circuit exactly because the open interaction chain embeds locally. For the MFIM ring, TAR-AVQDS uses 259 versus 300 naive CNOTs at $h_z = 0$, and 469 versus 574 at $h_z = 0.5$; depths are 151 versus 182 and 311 versus 368, respectively. Table 10 gives all results and audits. This extension requires no change to the tangent rule: only the graph object and its precomputed shortest paths differ.

Table 7: **Ansatz and routing decomposition for all eight required topology-aware circuits.** The physical CNOT column verifies $2N_{\theta,2q} + 3N_{\text{swap}}$.

Benchmark	topology	$N_{\theta,1q}$	$N_{\theta,2q}$	N_{θ}	logical CNOT	SWAP	physical CNOT
MFIM $h_z = 0$	A	60	66	126	132	0	132
MFIM $h_z = 0$	B	65	53	118	106	46	244
MFIM $h_z = 0.5$	A	168	116	284	232	0	232
MFIM $h_z = 0.5$	B	186	90	276	180	87	441
LSM $h_z = -0.7$	A	12	60	72	120	0	120
LSM $h_z = -0.7$	B	23	53	76	106	18	160
LSM $h_z = 1.6$	A	15	59	74	118	0	118
LSM $h_z = 1.6$	B	23	50	73	100	14	142

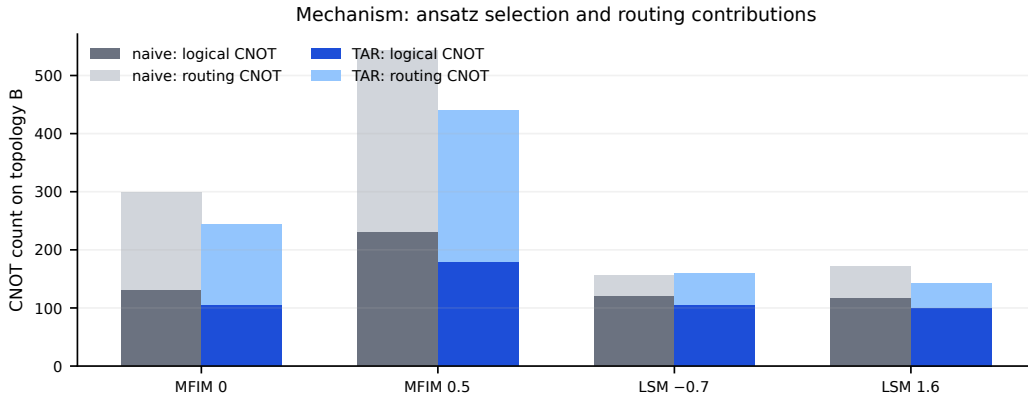


Figure 4: **CNOT mechanism on topology B.** Dark segments are the two CNOTs per selected two-qubit Pauli rotation; light segments are the three CNOTs per routed SWAP. TAR-AVQDS changes both terms because candidate choice changes the logical ansatz and persistent layout trajectory.

7 Limitations

The evidence is exact and complete for these eight-qubit tasks, but four limitations bound the conclusions. First, the study is noiseless. Additional SWAPs would be especially damaging under two-qubit noise, but hardware fidelity depends on calibration-specific edge error and duration; CNOT count and primitive depth are only transparent proxies. Error mitigation can alter this trade-off but does not remove its calibration dependence [46, 75, 23, 22]. Second, the Hamiltonian pools are one- and two-local. Higher-weight generators require Steiner-style parity networks, so “immediate route cost” would need a different exact definition; symmetry tapering changes the logical interaction graph again [12, 68].

Third, exhaustive initial-layout refinement is practical only because $8! = 40320$. It strengthens this comparison but is not a scalable subroutine. For larger systems, the same deterministic or stochastic allocator must be used for both smart naive routing and TAR-AVQDS, and the resulting margin should be reported with allocator variability. Fourth, the entangled LSM ground-state preparation circuit is excluded, following the original benchmark. This isolates dynamics but understates an end-to-end experiment’s depth; imaginary time and adaptive state-preparation methods are possible front ends [58, 52, 29, 28]. Finally, the single-step potential can be myopic, as the $h_z = -0.7$ result demonstrates; the method guarantees tangent admissibility and topology compliance, not global routing optimality.

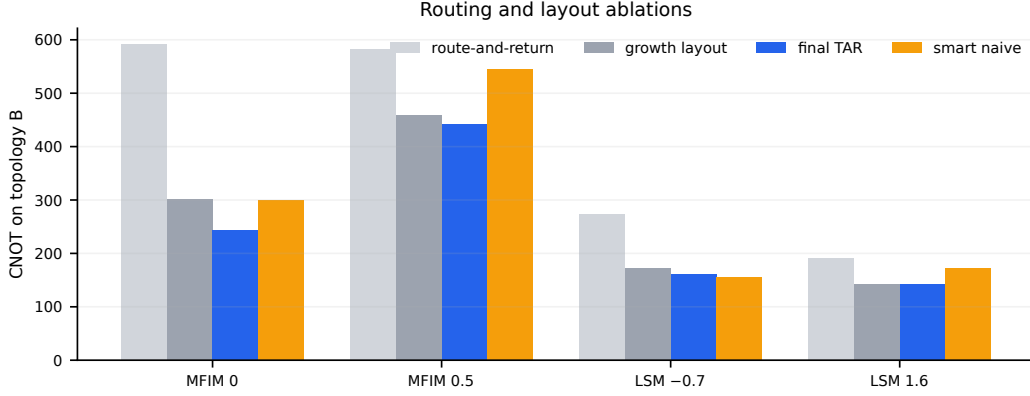


Figure 5: **Routing-component ablation on topology B.** Route-and-return removes persistent layout state; “growth layout” retains persistence but omits exhaustive final-layout refinement. Final TAR-AVQDS combines both components. Smart naive routes the unconstrained ansatz with the same exhaustive final-layout stage.

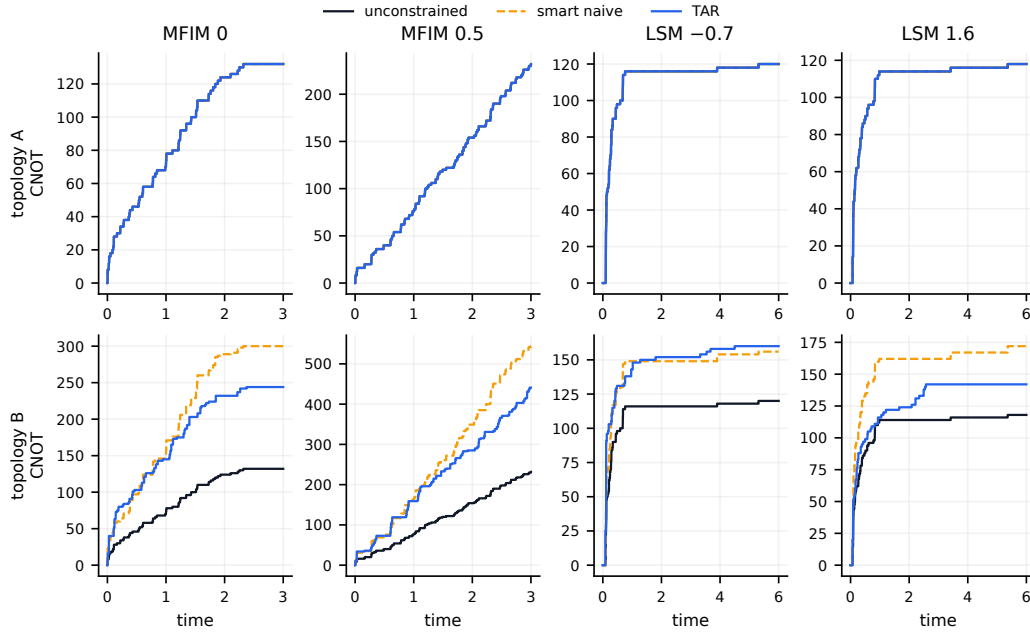


Figure 6: **Physical CNOT count through time for all eight benchmark–topology pairs and all three methods.** Each staircase is compiled from the full recorded gate prefix. Smart naive and unconstrained share a state trajectory but differ in routed count.

8 Observable-level validation

State fidelity is the acceptance metric, but it can obscure which physical quantities are reproduced. Figure 8 gives both MFIM Loschmidt echoes. Figure 9 gives energy and both nearest-neighbor XX and YY correlations for each LSM path. The archived arrays additionally contain X_0X_7 and Y_0Y_7 long-range correlations.

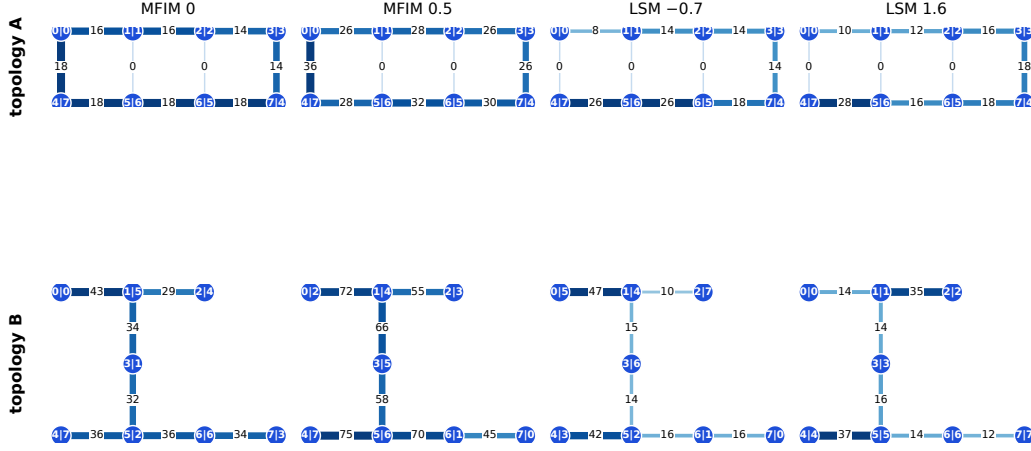


Figure 7: **Topology-compliant edge use and selected embeddings.** Rows are physical topologies and columns are dynamics. Edge labels are total two-qubit primitives and widths are normalized within each panel. A node label $p|\ell$ gives physical qubit p and the logical site ℓ initially placed there by exhaustive final-layout selection. No primitive occurs away from a drawn edge.

Table 8: **Adaptive-core wall time.** A/base and B/base compare each topology-aware run with the paired unconstrained run after adding the separate exhaustive-layout route time. “overrides” counts growth choices different from the original candidate rule. Exact-reference integration is excluded. All times are seconds except the two ratios.

Benchmark	base core	A core	A route	A total/base	B core	B route	B total/base	B overrides
MFIM $h_z = 0$	47.4	48.5	10.4	1.24	56.0	17.9	1.56	60
MFIM $h_z = 0.5$	138.1	124.3	16.0	1.02	120.5	30.3	1.09	164
LSM $h_z = -0.7$	48.4	48.5	8.7	1.18	69.8	11.4	1.68	29
LSM $h_z = 1.6$	55.1	54.2	8.4	1.14	63.3	10.3	1.34	25

9 Per-circuit compliance summary

Table 11 summarizes the eight compiled gate streams. “Two-qubit primitives” includes both rotation CNOTs and the three constituent CNOTs of every SWAP. Distinct edges counts which allowed physical edges were exercised; it is not a count of logical interaction bonds. Machine-readable archives additionally store the ordered physical pair of every primitive and its cumulative offset at every adaptive time step.

10 Calibration and graph diagnostics

Figure 10 visualizes the calibration behind Table 9. Figure 11 is independent of the dynamical run: it exhaustively finds the best static Hamiltonian-bond potential for each interaction graph/topology pair. This diagnostic explains why square and line LSM runs are negative controls while heavy-hex necessarily activates routing.

11 Reproducibility and integrity

All reported time series, final parameter arrays, operator sequences, growth decisions, mappings, and observables are saved under `proposal/logs/runs`. Every numerical archive has a SHA-256 digest in its run summary. Compliance archives under `proposal/logs/compliance` contain the declared edge array, ordered two-qubit pairs, gate sources, ansatz indices, growth steps, and per-step cumulative counts. The final audit verifies exact stop times, all fidelity floors, finite and monotone curves, CNOT identities, archive digests, declared edge equality, and zero off-edge pairs. The pairing

Table 9: **Declared one-problem calibration on MFIM $h_z = 0$, topology B.** Joint score is the geometric mean of CNOT and depth ratios to the smart naive circuit. Bold marks the frozen setting.

η	fidelity	N_θ	CNOT	depth	joint score
0.1	0.999396	118	244	148	0.782
0.2	0.999507	124	254	184	0.889
0.4	0.999422	122	255	170	0.856
0.6	0.999506	124	239	165	0.817

Table 10: **Optional linear-chain topology C.** The same $\eta = 0.1$, exact layout enumeration, compiler, and compliance audit are used.

Benchmark	base f	TAR f	naive CNOT	TAR CNOT	naive depth	TAR depth	N_θ	violations
MFIM $h_z = 0$	0.999494	0.999367	300	259	182	151	119	0
MFIM $h_z = 0.5$	0.999244	0.999094	574	469	368	311	277	0
LSM $h_z = -0.7$	0.999572	0.999572	120	120	191	191	72	0
LSM $h_z = 1.6$	0.999199	0.999199	118	118	190	190	74	0

manifest records the canonical-pool digest, initial-state digest, fixed integration configuration, time-grid digest, and selected embedding for every method and topology. Smart naive routing shares the unconstrained state and time grid exactly; TAR-AVQDS shares the initial state and adaptive-step rule, while its realized grid changes endogenously with its ansatz.

The source audit and exact reuse boundary are in proposal/codebase_notes.md; superseded strategies and every calibration run are in proposal/attempts_log.md. The executable entry points are run_experiment.py, run_authors_baseline.py, aggregate_and_plot.py, and audit_results.py. No pre-existing numerical result is used in the tables or figures. The numerical stack uses NumPy, SciPy, and QuTiP [32, 79, 37, 38]; measurement-aware deployment would additionally need to account for finite sampling and metric-estimation cost [20, 78].

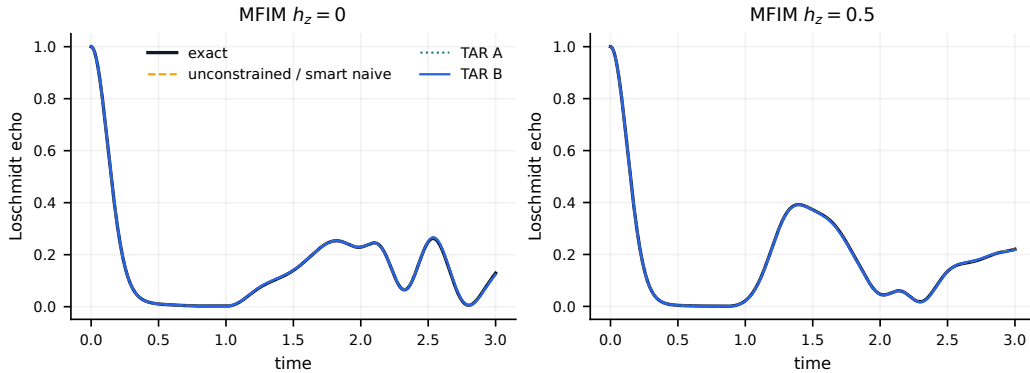


Figure 8: **Loschmidt echo for both MFIM quenches.** Exact, unconstrained/smart-naive, and both topology-aware trajectories are shown. Coincident curves reflect exact routing or the inert topology A selector.

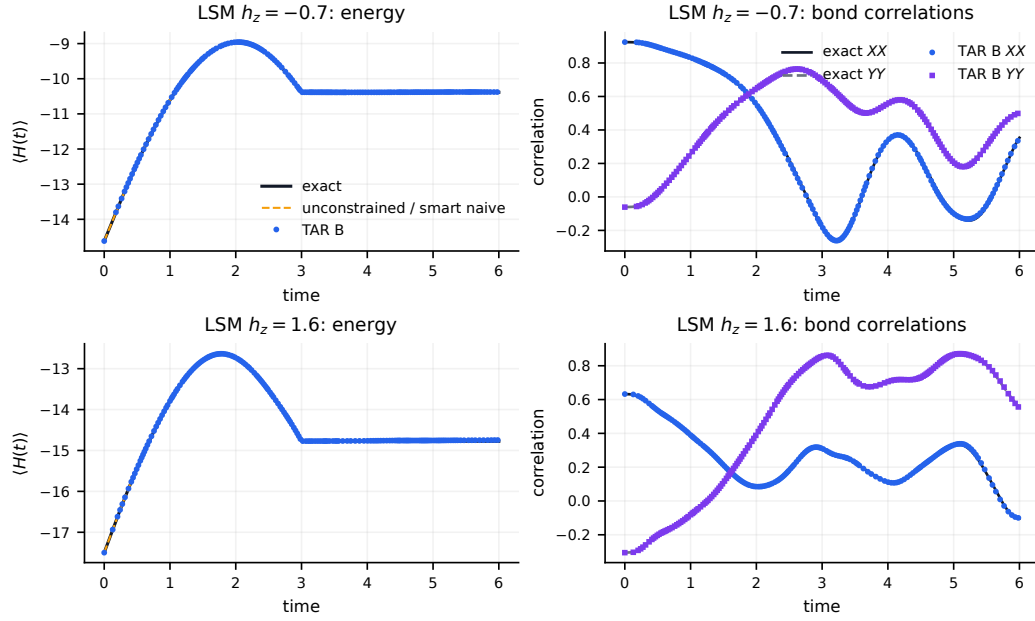


Figure 9: **Energy and spin correlations for both LSM ramps.** Markers subsample the recorded TAR-AVQDS topology B series only for visibility; the archives retain every adaptive step. Smart-naive observables coincide with unconstrained values.

Table 11: **Topology compliance for all eight TAR-AVQDS circuits.** Violations are independently recomputed from saved gate-pair arrays and the exact declared edge set.

Benchmark	topology	two-qubit primitives	distinct edges	violations
MFIM $h_z = 0$	A	132	8	0
MFIM $h_z = 0$	B	244	7	0
MFIM $h_z = 0.5$	A	232	8	0
MFIM $h_z = 0.5$	B	441	7	0
LSM $h_z = -0.7$	A	120	7	0
LSM $h_z = -0.7$	B	160	7	0
LSM $h_z = 1.6$	A	118	7	0
LSM $h_z = 1.6$	B	142	7	0

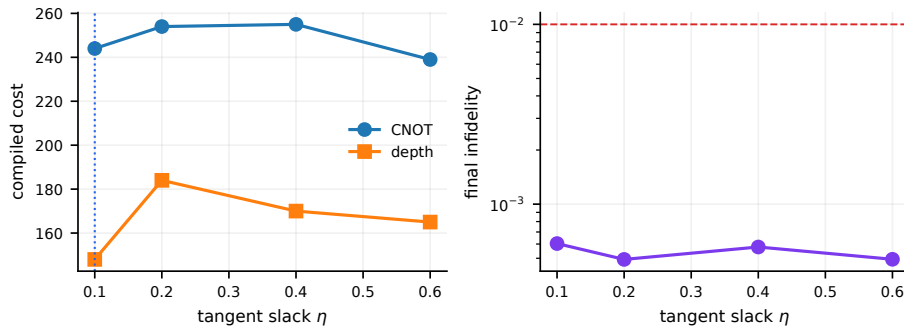


Figure 10: **Tangent-slack calibration.** The left panel shows the nonmonotonic count/depth trade-off; the right confirms that every candidate setting remains far inside the one-percent final-infidelity regime.

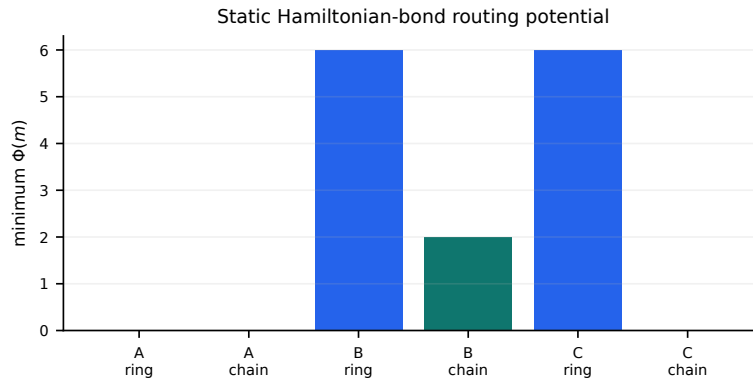


Figure 11: **Minimum static interaction potential.** Values are exact over all $8!$ logical-to-physical mappings. Zero means every ring or chain bond is simultaneously local.

12 Conclusion

TAR-AVQDS makes physical connectivity part of adaptive variational dynamics without changing McLachlan evolution. The method admits only near-best regularized tangent gains, then uses a persistent shortest-route state and Hamiltonian-bond potential to choose among them. Across every benchmark and topology it preserves the stated accuracy criterion and emits only allowed two-qubit gates. It is exactly inert on a square embedding that needs no routing, while on heavy-hex it reduces mean smart-naïve CNOT and depth costs by 13.1% and 16.5%. The single CNOT-count underperformance is informative rather than hidden: local admissible choices can improve logical size and depth while worsening the future layout. These results support a broader design principle for adaptive quantum algorithms: hardware should choose within a variationally certified set, rather than replace the physical objective that defines the set. Extending that principle to nonequilibrium embedding solvers is a natural direction [67, 5, 43, 82].

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